MILITARY SPECIFICATION SHEET

WAVEGUIDE ASSEMBLIES, FLEXIBLE, TWISTABLE, CLASS 5

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the waveguide assemblies described herein shall consist of this document and the latest issue of Specification MIL-W-287.

REQUIREMENTS: See table I.

Part number: See table I.

Extent of qualification: Qualification obtained for one assembly shall constitute qualification for other assemblies as specified in table II.

Substitution data: See table III.

FSC 5985

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TABLE I. Flexible, twistable waveguide assemblies.

Group number	1/ Part number M28B7/6-	TE_{10} mode frequency range (GHz)	Minimum center line bending radius (inch)		Repeated twist (degrees/ft) 1x10⁵ cycles	Insertion loss (dB/ft)	Axial twist (degrees/ft)	Pressurization lb_f/in^2 gage	2/ VSWR max at			
			E plane	H plane					12°	24°	36°	>36°
1	01-XXXX	3.5-8.2	3.5	7.0	±10°	0.10	90°	30	1.20	1.25	1.30	1.38
	02-XXXX	4.75-11.0	3.0	6.0	±10°	0.15	90°	35	1.20	1.25	1.30	1.38
	03-XXXX	7.5-18.0	2.0	4.0	±10°	0.20	90°	45	1.20	1.25	1.30	1.38

Flexure (degrees/ft) 1x10⁵ cycles		Jacket material	Retention force (lbf)	Oper temp range -55 to	Mates with standard ridge waveguide MIL-23351/4-	Flanges		
E plane	H plane					Flange 1 MIL-39000( )	Flange 2 MIL-39000	Material 3/
30	15	Neoprene	N/A	+100°C	029	3-058 (gasket)	3-032 (cover)	AL
30	15	Neoprene	N/A	+100°C	033	3-070 (gasket)	3-044 (cover)	AL
30	15	Neoprene	N/A	+100°C	037	3-082 (gasket)	3-074 (cover)	AL

1/ The complete part number shall be the number shown and a four-digit number indicating the nominal relaxed length in inches.
2/ The VSWR value for any length between the lengths specified shall be the next higher value (e.g., for a 15 inch length of M28B7/6-02, the VSWR would be 1.25:1).
3/ AL = Aluminum alloy.

TABLE II. Grouping for qualification.

Group number	Qualifying dash number	Dash numbers of assemblies qualified
1	02	01 and 03

TABLE III. Substitution data. 1/

Dash number	Manufacturer's code (FSCM	Manufacturer's part number
01	87864	TDFT-29019
02	87864	TDFT-22009
03	87864	TDFT-14007 series

1/ Assemblies covered by this specification sheet are substitutable for the manufacturer's part numbers shown. This information in no way implies that the manufacturer's part is substitutable for the military part.

Custodians
Army - ER
Navy - EC
Air Force - 85

Review activities
Army - AV, MI
Navy - OS
Air Force - 17, 99
DLA - ES

User activities
Army - AR
Navy - MC, CG, AS
Air Force - 11, 19

Preparing activity,
Navy - EC

Agent
DLA - ES

Project 5905-0919-4

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DOCUMENT IDENTIFIER AND TITLE: MIL-W-287/6 - Waveguide Assemblies, Flexible, Twistable, Class 5

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
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   A. GIVE PARAGRAPH NUMBER AND WORDING

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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   ☐ YES ☐ NO (If "Yes", in what way?)

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